

Project synopsis

HLD Navigator - AUTOSAR HLD Document Analysis Assistant

Case Study 1 | TechPulse FY-26 Applied AI/ML

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Proposed work and delivered scope

Build a local AUTOSAR HLD review assistant that turns supported document statements into evidence-linked architecture proposals, permits human correction/approval, retrieves cited context, compares revisions and exports reviewed component/dependency reports.

Delivered software includes PDF/Markdown/text ingestion, optional real OCR, governed review, workspace access controls, FTS search, optional learned embeddings/local answers, architecture findings, dependency visualization, revision impact analysis, recovery tools and a reproducible fictional interview demo.

Method and evaluation

Use explicit extraction rules, immutable source bytes/locations, separate review states, lexical and optional dense retrieval, and structured answer citation validation. Evaluate with regressions, real HTTP/UI integration, frozen public documents, real OCR and fixed synthetic answer cases.

The final local regression suite passes 499 tests. Retrieval and public extraction results expose remaining limitations, rather than establishing OEM reliability. No independent automotive validation or measured time saving has occurred.

Expected deliverables

Source and lockfile, synthetic/public input corpus with provenance, model/prompt configuration, evaluation evidence, technical report, editable presentation, demo instructions and unsigned integrity/AI-use declarations.

Faculty approval

This synopsis is a prepared draft. Dr. D. Palmani has not supplied approval or a signature.

Faculty signature: _____ Date: _____

Student signature: _____ Register/team ID: _____